

1. EXECUTIVE SUMMARY

On 16 June 2026 at 20:25 UTC, the Wazuh SIEM platform detected a brute force attack targeting the Windows endpoint at `192.168.56.20`. The attack originated from `192.168.56.103` and targeted the administrator account via SMB port 445. The attack was detected and contained with **no successful unauthorized access**.

2. ATTACK TIMELINE

All times recorded in UTC.

TIME (UTC)	EVENT
20:17	Nmap reconnaissance scan detected from 192.168.56.103
20:25	Brute force attack commenced using Metasploit smb_login module
20:25	Wazuh Rule 60122 triggered — Severity Level 10 (HIGH)
20:25	11 authentication failures recorded against administrator account
20:26	Administrator account locked out — attack automatically failed
20:30	SOC Analyst investigated and confirmed attack; containment initiated

3. INDICATORS OF COMPROMISE (IOCs)

Source IP	192.168.56.103 (Kali Linux attacker)
Target IP	192.168.56.20 (Windows endpoint)
Target Account	administrator
Attack Port	445 / TCP (SMB)
Tool Identified	Metasploit Framework – smb_login module
Windows Event IDs	4625 (Failed Login) 4740 (Account Lockout)
Wazuh Alert Rule	60122

4. MITRE ATT&CK MAPPING

Tactic	TA0006 — Credential Access
Technique	T1110 — Brute Force
Sub-technique	T1110.001 — Password Guessing

5. INVESTIGATION FINDINGS

- Total failed login attempts: 11
- Attack duration: approximately 5 minutes (20:25 – 20:30 UTC)
- No successful login was achieved
- Account lockout policy activated correctly, automatically terminating the attack
- No lateral movement detected within the network
- No data exfiltration detected

6. CONTAINMENT ACTIONS TAKEN

- Source IP 192.168.56.103 blocked via firewall rule
- Administrator account unlocked following investigation confirmation
- Password reset performed on the targeted administrator account

7. ROOT CAUSE ANALYSIS

SMB port 445 was exposed to the network without IP-based access restrictions, allowing a remote attacker to initiate unrestricted brute force authentication attempts against the Windows endpoint. The absence of IP allowlisting on SMB is the primary contributing factor to this incident.

8. RECOMMENDATIONS

1. Implement account lockout after 3 failed login attempts (current threshold is too permissive)
2. Restrict SMB port 445 access to trusted and explicitly approved IP ranges only
3. Implement Multi-Factor Authentication (MFA) on all privileged and administrator accounts
4. Deploy network segmentation to limit the blast radius of potential lateral movement
5. Enable Wazuh Active Response to automatically block attacking IP addresses in real time

9. LESSONS LEARNED

The Wazuh SIEM successfully detected the attack in real time and provided complete forensic data sufficient to conduct a thorough post-incident investigation. The account lockout policy prevented any successful unauthorised access, clearly demonstrating the value of baseline preventive security controls.

This incident highlights the critical importance of network-level access controls and proactive hardening measures aligned with the ASD Essential Eight Maturity Model. The SOC lab environment has validated its detection capability; the next maturity step is automating the response.

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